

# SHIFTWM: WORLD MODELS THAT MOVE WHAT THEY SAW

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## ABSTRACT

Latent world models forecast in the feature space of a frozen visual foundation model, but at every future step they regenerate the entire feature grid, even though most of a manipulation scene only moves. We introduce SHIFTWM, a world model that forecasts by *moving what it has already seen*. For each prefix of future actions, SHIFTWM predicts a local soft transport of observed patch features, a gate that decides where transport is used, and a correction for content that transport cannot explain. We show that the optimal forecast under motion is itself such a transport, that additive heads must instead learn the full feature contrast along the motion, and that anchoring every horizon to measured features prevents compounding error. On held-out real robot video (DROID), SHIFTWM removes 28.0% of persistence error against 17.5% for a matched recursive predictor, and it is the best predictor on seven surgical robot tasks. SHIFTWM also works as a plug-in head for published world models: added to V-JEPA 2-AC and fine-tuned under an identical budget, it lowers forecast error by 15.7% (24.6%  $\rightarrow$  36.4% skill); added to DINO-WM with its official code, it changes latent prediction error by  $-\text{pending}\%$  (PushT) and  $-\text{pending}\%$  (Wall). The learned transport field and gate are explicit, inspectable tensors that show where the model predicts motion.

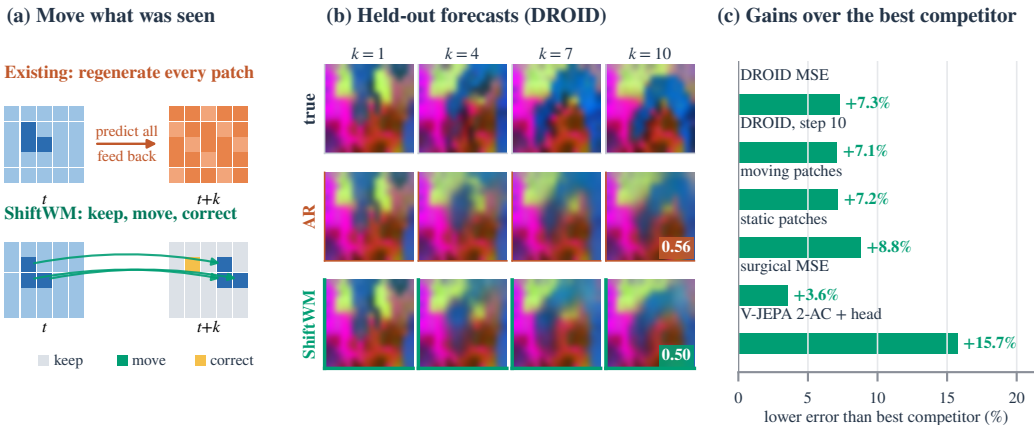


Figure 1: **SHIFTWM moves what it has seen instead of regenerating it.** (a) Recursive world models regenerate every patch and feed back their forecasts; SHIFTWM keeps, moves and corrects observed features. (b) Held-out DROID forecasts (feature maps, step-10 error in the corner). (c) Error reduction over the strongest competitor per metric and benchmark.

## 1 INTRODUCTION

A world model predicts how observations change in response to actions. Recent work shows that forecasting in the feature space of a frozen, self-supervised encoder such as DINOv2 or V-JEPA 2 gives compact models that support planning without generating pixels (Zhou et al., 2025; Assran et al., 2025; Maes et al., 2026b; Sobal et al., 2025). These predictors are *generative in feature space*.

At every future step they output the complete grid of patch features, usually by feeding their own predictions back as inputs. In manipulation and surgical video, however, most of the scene persists: tissue, tools, the table and the robot move, but they are rarely created or destroyed. Regenerating persistent content at every step wastes capacity and lets errors compound under recursion.

We ask a simple question. *Can a latent world model forecast by transporting features it has already observed, and synthesise only what transport cannot explain?* Motion-based video prediction answered a version of this question in pixel space (Finn et al., 2016; Ebert et al., 2017; De Brabandere et al., 2016; Liu et al., 2017). The idea has not been tested for foundation-feature world models used for planning, where features are semantic, high-dimensional and coarse in space. Three questions are open: whether transport helps once the predictor is strong, whether it survives held-out real data, and what it costs for control.

We introduce SHIFTWm (Figures 1 and 2). A memory encoder reads the patch features of the last few observed frames. For each future step, a decoder conditioned on the causal action-prefix state produces per-patch queries. Each query (i) attends to a local window of *observed* feature locations and copies their content, (ii) gates between staying in place and moving, and (iii) adds a learned correction. Every horizon is predicted directly from measured features, so errors do not compound through recursion, and the transport weights form a motion field that we can inspect and compare against optical flow.

**What is new.** SHIFTWm changes *what* a latent world model predicts. Instead of regressing the next feature grid, it predicts, for every future step, a motion of the grid it has already observed: a per-patch transport over measured features, a gate and a residual. Three properties distinguish it from prior work. (i) Unlike pixel-warping predictors (Finn et al., 2016; Ebert et al., 2017; Liu et al., 2017), it moves *semantic* foundation features that planners consume directly, and the move is conditioned on an explicit action prefix. (ii) Unlike recursive latent world models (Zhou et al., 2025; Maes et al., 2026b; Assran et al., 2025), every horizon is computed directly from measured features, so errors cannot compound through feedback. (iii) Its intermediate quantities—where each patch comes from, whether it moves, and what is new—are explicit tensors that can be inspected and measured (Figure 10).

## Contributions.

- **Method.** To our knowledge, SHIFTWm is the first latent world model that forecasts by action-conditioned transport of frozen foundation features. It is a drop-in output head for any frozen patch encoder and adds  $< 0.3\%$  parameters to a matched predictor.
- **Theory.** Under a motion model, the MSE-optimal forecast is a transport of observed features; additive heads must learn the feature contrast along the motion; recursive rollouts compound error (Section 3.1).
- **Held-out evidence.** Against predictors with the same backbone, SHIFTWm has the lowest held-out error on real robot video (DROID; 28.0% vs. 17.5% skill for recursive AR) and on seven surgical robot tasks, and we evaluate it on IWS, BridgeData V2, RT-1 and three planning suites.
- **Plug-in for published world models.** Added to V-JEPA 2-AC, the head lowers forecast error by 15.7% under an identical fine-tuning budget; added to DINO-WM with its official code, it changes latent prediction error by  $-\text{pending}\%$  (PushT) and  $-\text{pending}\%$  (Wall) (Section 4.3).
- **Mechanism.** We show that transport keeps forecasts sharp where regression blurs, that the learned field follows measured motion, and how an action-contrastive objective keeps anchored forecasts sensitive to actions for planning.

## 2 RELATED WORK

**World models in foundation-feature and JEPA latent spaces.** A recent line of work predicts dynamics in the embedding space of a pretrained or jointly learned encoder instead of in pixels. DINO-WM trains a causal ViT predictor on frozen DINOv2 patch features and plans with model-predictive control (Zhou et al., 2025; Oquab et al., 2024); PLDM, LeWM and JEPA-WMs study joint-embedding predictive models learned from offline trajectories (Sobal et al., 2025; Maes et al., 2026b; Balestriero & LeCun, 2025; Terver et al., 2026), and V-JEPA 2-AC post-trains an action-

conditioned predictor on DROID for zero-shot robot planning (Assran et al., 2025; Khazatsky et al., 2024). DINO-world, DINO-Foresight, VFMF and FlowWM forecast future foundation features, deterministically or generatively, and evaluate them through downstream dense heads (Baldassarre et al., 2025; Karypidis et al., 2025; Boduljak et al., 2025; Porcher et al., 2026). DDP-WM separates sparse dynamic regions from a static background, and RLA-WM models residual latent actions over visual features (Yin et al., 2026; Zhang et al., 2026a). Complementary analyses ask which latent spaces and objectives make predictions action-relevant (Nilaksh et al., 2026; Yeom et al., 2026; Zhang et al., 2026b; Huang et al., 2026). In these models the predictor synthesizes each future feature map and, in most cases, conditions later steps on its own outputs. ShiftWM keeps the same frozen-encoder interface, applicable to DINOv2 or DINOv3 features (Siméoni et al., 2025), but changes the output parameterization: every forecast is expressed as a transport of observed features plus a correction, so the predictor is never required to regenerate static content.

**Generative world models and model-based reinforcement learning.** Pixel-space generative world models learn high-fidelity, action-controllable video simulators. NWM conditions a diffusion transformer on navigation actions (Bar et al., 2025); Cosmos and Cosmos-Predict2.5 provide large video foundation models for physical AI (NVIDIA, 2025a;b) that Cosmos Policy adapts for control (Kim et al., 2026); Ctrl-World, IRASim, UniSim and IWS target robot manipulation and policy evaluation (Guo et al., 2026b; Zhu et al., 2025; Yang et al., 2024; Wang et al., 2026); and the Genie family generates interactive environments from video, image or text prompts (Bruce et al., 2024; Parker-Holder & Fruchter, 2025). Model-based reinforcement learning instead learns task-specific latent dynamics jointly with rewards and policies, from PlaNet and DreamerV3 to TD-MPC2, IRIS and DIAMOND (Hafner et al., 2019; 2025; Hansen et al., 2024; Micheli et al., 2023; Alonso et al., 2024). These systems optimize sample fidelity or online return with their own encoders, data and, often, billions of parameters, so they do not share our contract of a fixed frozen encoder, identical observations and actions, and targets in common feature coordinates. We therefore treat them as complementary rather than as compute-matched baselines: they address rendering fidelity and online control, whereas we isolate the output parameterization of a compact predictor on a fixed representation.

**Transformation- and transport-based prediction.** Predicting the future by moving existing content has a long history in video prediction. CDNA and DNA predict action-conditioned convolutional kernels or pixel-motion distributions that transform the previous frame (Finn et al., 2016), dynamic filter networks generate input-dependent filters (De Brabandere et al., 2016), and Vondrick and Torralba transform past pixels with adversarially trained transformers (Vondrick & Torralba, 2017). SNA adds temporal skip connections to the first observed image to handle occlusion (Ebert et al., 2017), while flow-warping approaches synthesize frames by resampling along predicted voxel flow (Liu et al., 2017); SAVP and MCVD extend pixel prediction with stochastic and diffusion models (Lee et al., 2018; Voleti et al., 2022). Recent robot world models reintroduce explicit motion: FlowDreamer predicts 3D scene flow before diffusion rendering (Guo et al., 2026a), AHEAD forecasts VLA patch tokens conditioned on optical-flow velocity (Syed et al., 2026), and anchored navigation world models resist drift through anchor frames and epipolar guidance (Luan et al., 2026). Object-centric models such as SlotFormer and SOLD impose structure through slots (Wu et al., 2023; Mosbach et al., 2025). ShiftWM differs from these designs in four respects. It transports frozen foundation features rather than pixels; its source is a fixed memory of *observed* feature grids that remains unchanged at every horizon, so predicted states are never fed back as inputs; each horizon- $h$  forecast is produced directly from the action prefix  $a_{1:h}$ ; and the transport is a local softmax over a spatial neighborhood, combined with a gated identity path and a learned correction for disocclusion and appearance change.

**World models for surgery and evaluation suites.** Surgical world modeling is driven by generative video models: Endora simulates endoscopy video (Li et al., 2024), SurgWM learns latent-action-controllable surgical video (Koju et al., 2025), SurgVista targets long-horizon instrument-tissue dynamics (Pan et al., 2026), SWoMo combines neural and symbolic components for cataract surgery (Sivakumar et al., 2026), and Cosmos-H-Surgical learns surgical robot policies from videos through world modeling (He et al., 2025). Open-H-Embodiment releases paired video and kinematics across surgical platforms, together with a kinematics-conditioned surgical simulator (Open-H-Embodiment Consortium, 2026); SutureBot, JIGSAWS, Cholec80 and CholecT50 provide surgical

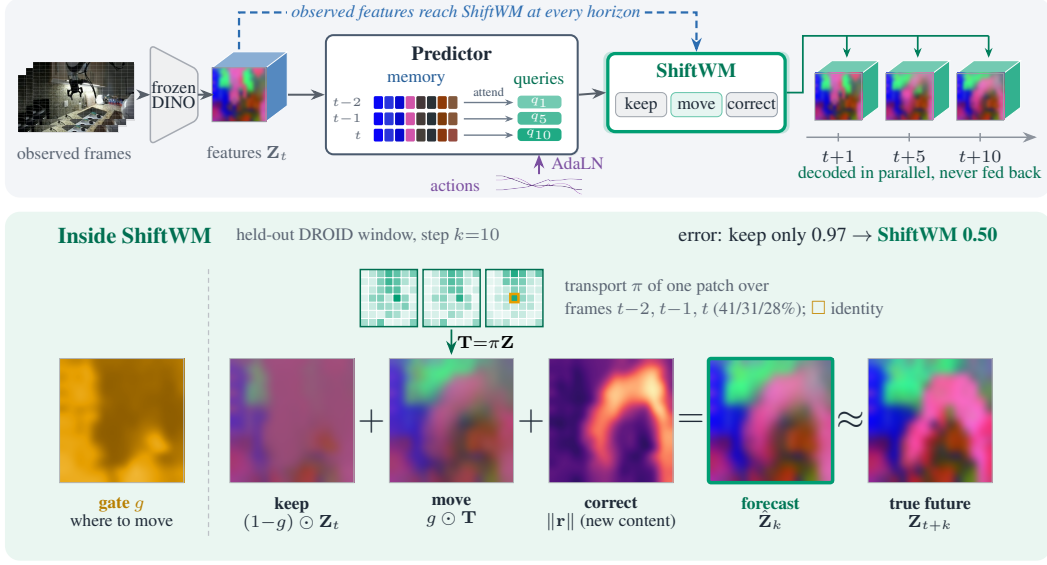


Figure 2: **SHIFTWM architecture.** Top: a predictor turns observed features and actions into one query per patch and step; SHIFTWM decodes all steps in parallel from observed features. Bottom: on a held-out window, each forecast is kept + moved + corrected observed features.

demonstrations and task, skill and workflow annotations (Haworth et al., 2025; Gao et al., 2014; Twinanda et al., 2017; Nwoye et al., 2022), and SurgWMBench benchmarks instrument-motion planning (Liu et al., 2026). We use Open-H dVRK recordings to test kinematics-conditioned forecasting in feature space, a regime these pixel generators do not target. More broadly, robot datasets such as DROID, BridgeData V2 and Open X-Embodiment supply action-labeled video (Khazatsky et al., 2024; Walke et al., 2023; Open X-Embodiment Collaboration, 2024). WorldScore, World-ModelBench, EWMBench and Physics-IQ score generated video for controllability and physical plausibility (Duan et al., 2025; Li et al., 2025; Yue et al., 2025; Motamed et al., 2026), whereas stable-worldmodel standardizes latent planning evaluation (Maes et al., 2026a). We follow the latter protocols for control and report feature-space forecasting error for prediction.

### 3 METHOD

Figure 2 summarises the model. Its lower row shows the actual tensors of a trained model on a held-out DROID window: each term of Equation (2) as a feature map (a shared PCA projection to colour), the learned gate, and the transport weights of one patch.

**Problem.** A frozen encoder  $E$  maps an image to a grid of  $N = G \times G$  patch features  $\mathbf{Z}_t \in \mathbb{R}^{N \times C}$ . Given  $H$  observed grids  $\mathbf{Z}_{-H+1:0}$ , the actions between them  $\mathbf{a}_{-H+1:-1}$ , and a future action sequence  $\mathbf{a}_{0:K-1}$ , a world model predicts  $\hat{\mathbf{Z}}_{1:K}$ . The forecast  $\hat{\mathbf{Z}}_k$  may depend only on  $\mathbf{a}_{<k}$  (causality). Features are standardised per channel with train-split statistics that are shared across positions.

**Memory encoder.** Each observed patch feature is projected to width  $d$  and receives a spatial embedding, a frame embedding and an embedding of the action applied after that frame.  $L_e$  transformer layers over all  $H \cdot N$  tokens give memory states  $\mathbf{M} \in \mathbb{R}^{H \times N \times d}$ .

**Action-prefix queries.** A unidirectional GRU over embedded actions gives a causal prefix state. We add a horizon embedding to form  $\mathbf{c}_k$ . Query tokens start from the memory states of the last frame, and  $L_d$  decoder layers apply self-attention across patches and cross-attention to  $\mathbf{M}$ . Each layer is modulated by  $\mathbf{c}_k$  through adaptive layer normalisation with zero-initialised gains (Peebles & Xie, 2023). The result is  $\mathbf{h}_{k,i} \in \mathbb{R}^d$  for patch  $i$  at step  $k$ . All  $K$  steps are decoded in parallel.

**Local feature transport.** For query patch  $i$ , let  $\mathcal{W}(i)$  be the  $w \times w$  spatial window around  $i$  in each of the last  $S \leq H$  observed frames. Transport weights are

$$\pi_{k,i}(j) = \text{softmax}_{j \in \mathcal{W}(i)} \left( \frac{1}{\sqrt{d_k}} (\mathbf{W}_q \mathbf{h}_{k,i})^\top (\mathbf{W}_k \mathbf{M}_j) + \beta \mathbb{I}[j = (0, i)] \right), \quad (1)$$

where  $\beta$  is a learned identity bias toward the same location in the last frame. Transport moves *measured* features, not latent states:  $\mathbf{T}_{k,i} = \sum_j \pi_{k,i}(j) \mathbf{Z}_j$ .

**Gated output.** A per-patch gate  $g_{k,i} = \sigma(\mathbf{w}_g^\top \mathbf{h}_{k,i} + b_g)$  and a correction  $\mathbf{r}_{k,i} = \mathbf{s} \odot \mathbf{W}_o \mathbf{h}_{k,i}$  give

$$\hat{\mathbf{Z}}_{k,i} = (1 - g_{k,i}) \mathbf{Z}_{0,i} + g_{k,i} \mathbf{T}_{k,i} + \mathbf{r}_{k,i}. \quad (2)$$

$\mathbf{W}_o$  is zero-initialised and  $b_g = -3$ , so an untrained model starts close to persistence and learns departures from it.  $\mathbf{s} \in \mathbb{R}^C$  is a learned per-channel scale. Appendix D.2 compares it with the bounded (tanh) correction used in earlier variants.

**Training.** We minimise the mean squared error over all horizons,  $\mathcal{L}_{\text{pred}} = \frac{1}{K} \sum_k \|\hat{\mathbf{Z}}_k - \mathbf{Z}_k\|^2$ . Anchoring every forecast to observed features makes it easy to ignore actions: a forecast that stays close to  $\mathbf{Z}_0$  scores well on average. For planning we therefore add an action-contrastive hinge. For each sample it compares the prediction error under the true future actions with the error under actions borrowed from another sample in the batch:

$$\mathcal{L}_{\text{act}} = \max(0, m - (\|\hat{\mathbf{Z}}(\tilde{\mathbf{a}}) - \mathbf{Z}\|^2 - \text{sg} \|\hat{\mathbf{Z}}(\mathbf{a}) - \mathbf{Z}\|^2)), \quad \mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{act}}. \quad (3)$$

### 3.1 WHY TRANSPORT: A MOTION VIEW OF FEATURE FORECASTING

We make the inductive bias of Equation (2) precise under a simple generative model; Figure 3 shows the geometry for one patch. Write the observed grid as  $\mathbf{Z}_0 \in \mathbb{R}^{N \times C}$  and suppose the grid at step  $k$  arises from motion plus innovation,

$$\mathbf{Z}_k = \mathbf{P}_k \mathbf{Z}_0 + \boldsymbol{\varepsilon}_k, \quad (4)$$

where  $\mathbf{P}_k \in \{0, 1\}^{N \times N}$  is a row-stochastic *motion* matrix (each patch copies one source patch;  $\mathbf{P}_k = \mathbf{I}$  is a static scene) and  $\boldsymbol{\varepsilon}_k$  is new content with  $\mathbb{E}[\boldsymbol{\varepsilon}_k \mid \mathbf{P}_k, \mathbf{Z}_0, \mathbf{a}] = \mathbf{0}$ . Motion is uncertain given the context  $c = (\mathbf{Z}_{\leq 0}, \mathbf{a}_{< k})$ .

**Proposition 1** (The optimal forecast is a transport). *Under Equation (4), the MSE-optimal forecast is  $\mathbb{E}[\mathbf{Z}_k \mid c] = \bar{\mathbf{P}}_k \mathbf{Z}_0$  with  $\bar{\mathbf{P}}_k = \mathbb{E}[\mathbf{P}_k \mid c]$ , a row-stochastic matrix. If the support of each row of  $\mathbf{P}_k$  lies in the transport window  $\mathcal{W}$ , then  $\bar{\mathbf{P}}_k \mathbf{Z}_0$  can be approximated to arbitrary precision by the transport of Equation (1) with  $g=1$  and  $\mathbf{r}=\mathbf{0}$ , given logits that the decoder can express.*

The optimal forecast therefore never leaves the convex hull of *observed* features: it is a probability-weighted mixture of where each patch may have come from. SHIFTWM represents it with a bounded softmax over observed features and needs no correction for moved content. An additive head  $\mathbf{Z}_0 + \mathbf{W}_o \mathbf{h}$  must instead synthesise the whole displacement field in feature space:

**Proposition 2** (Cost of regenerating motion). *For a deterministic motion  $\mathbf{P}$  (a patch permutation), the residual an additive head must output is  $\mathbf{R} = (\mathbf{P} - \mathbf{I})\mathbf{Z}_0$  with  $\|\mathbf{R}\|_F^2 = \sum_i \|\mathbf{z}_{0,\pi(i)} - \mathbf{z}_{0,i}\|^2$ , the feature contrast along the motion. The residual required by SHIFTWM is  $\mathbf{0}$  for every such  $\mathbf{P}$  within the window.*

Proposition 2 explains where the gains concentrate: the residual grows with both motion and feature contrast, which is largest at object boundaries of the moving arm and instruments (Table 16). On

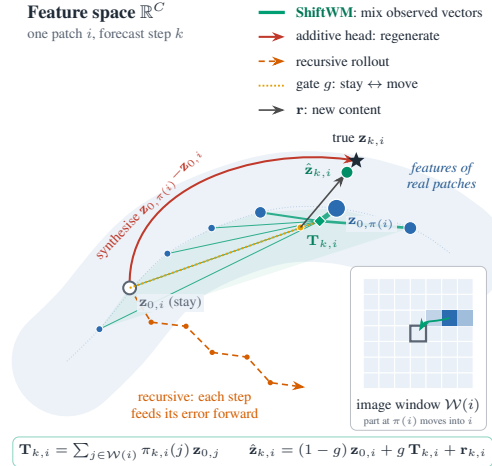


Figure 3: **One patch in feature space.** SHIFTWM mixes observed vectors; additive heads regenerate the displacement; recursion feeds its error forward.

static patches ( $\mathbf{P} = \mathbf{I}$ ) the transport identity path is exact, whereas a regression head must learn to output zero there, and any approximation error corrupts the static scene; we observe exactly this in Table 16. Finally, SHIFTWLM predicts every horizon from measured features, so errors cannot compound through feedback:

**Proposition 3** (No compounding). *Let a recursive predictor  $f$  be  $L$ -Lipschitz in its input with one-step error at most  $\delta$ . Its  $k$ -step rollout error is bounded by  $\delta \sum_{j=0}^{k-1} L^j$ , which grows geometrically for  $L > 1$ . A direct predictor with per-horizon error  $\delta_k$  incurs exactly  $\delta_k$ , independent of its errors at other horizons.*

Proofs are in Appendix B.2.

**Matched baselines.** To isolate the output head, all learned predictors share the same memory encoder, decoder, width and depth. **Direct** replaces Equation (2) with  $\mathbf{Z}_0 + \mathbf{W}_o \mathbf{h}_k$ . It has the same cross-patch attention, so it controls for spatial communication. **AR** predicts a residual to the most recent grid and is rolled out recursively, with gradients through the rollout. **AR-TF** follows DINO-WM: it is trained with teacher forcing to predict absolute features and is rolled out at test time. **Persistence** copies  $\mathbf{Z}_0$ , and **Linear** extrapolates the last feature difference.

**Planning.** For control we plug each predictor into the same model-predictive planner (CEM) with the official budgets of each benchmark. The cost is the feature distance between the predicted grid and the goal grid (Section 4.5).

## 4 EXPERIMENTS

We ask four questions. **(Q1)** Does transport improve held-out forecasting over matched predictors on real robot and surgical video (Section 4.2)? **(Q2)** Does the SHIFTWLM head improve strong *published* world models when plugged into them (Section 4.3)? **(Q3)** Which components matter (Section 4.4), and what happens in closed-loop planning (Section 4.5)? **(Q4)** Why does it work (Section 5)? Selection uses validation splits only; test splits are scored once.

### 4.1 SETUP

**Datasets** (Table 5). **DROID** (Khazatsky et al., 2024): 1,126 real Franka episodes from 24 official shards, split by recording session (851/143/132). We train on the primary exterior camera and evaluate transfer to the second exterior camera. One model step is 5 frames (0.33 s) with a 35-D block of Cartesian and gripper commands; we forecast  $K=10$  steps (3.3 s). **Open-H-Embodiment, Hamlyn dVRK** (Open-H-Embodiment Consortium, 2026): 1,013 episodes of seven surgical training tasks, 80-D bimanual Cartesian action blocks, episode-disjoint splits per task (708/152/153). **IWS** (Wang et al., 2026; Zhang et al., 2026a): single-image forecasting on PushT, Box and Rope with the official RLA-WM handles. **BridgeData V2** (Walke et al., 2023) and **RT-1 (Fractal)** (Open X-Embodiment Collaboration, 2024): pinned Open-X subsets (734 and 10,700 episodes). **Planning:** PushT, TwoRoom and Reacher with the official LeWM/stable-worldmodel protocol (Maes et al., 2026b;a); PushT and Wall with the official DINO-WM protocol.

**Models and metrics.** All learned predictors share a frozen DINOv2-S/14 encoder ( $16 \times 16$  patches, no pooling), width 384 and  $L_e=L_d=4$  ( $\approx 22$ M parameters), and the same optimiser, schedule, steps and batch size (Appendix B.1); they differ only in the output head and rollout. The primary metric is standardised feature MSE on held-out episodes, averaged over horizons and at the final step. We also report *skill*, the fraction of persistence error a model removes; being scale-free, it compares models in different feature spaces. Episodes are the statistical unit; we report means over seeds and paired session-bootstrap 95% intervals (Appendix C.5).

### 4.2 HELD-OUT FORECASTING ACROSS ROBOTS AND SURGERY (Q1)

Table 1 compares SHIFTWLM with predictors that share its backbone and differ only in how the forecast is formed. On held-out DROID, SHIFTWLM achieves the lowest error of all methods: it removes 28.0% of the persistence error, against 22.3% for Direct and 17.5% for AR, i.e. 7.3% lower

Table 1: **Held-out forecasting error** (feature MSE ↓, mean of 3 seeds). **Best, second**; †: paired 95% CI vs. best baseline excludes 0. [Interim: seeds completed – DROID: ar 2/3, ar-tf 2/3, direct 2/3, shiftwm 2/3; Hamlyn: ar 0/3, ar-tf 0/3, direct 0/3, shiftwm 0/3.]

| Method                            | DROID         |               | DROID cam 2 | Open-H        | IWS     | Bridge  | RT-1      |
|-----------------------------------|---------------|---------------|-------------|---------------|---------|---------|-----------|
|                                   | avg           | $k=10$        | (zero-shot) | surgical      |         | V2      | (Fractal) |
| Persistence                       | 0.249         | 0.326         | pending     | 0.444         | pending | pending | pending   |
| Linear extrapolation              | 4.812         | 11.855        | pending     | 10.824        | pending | pending | pending   |
| AR-TF (DINO-WM-style)             | 0.270         | 0.400         | pending     | 0.285         | pending | pending | pending   |
| AR (rollout-trained)              | 0.206         | 0.264         | pending     | 0.235         | pending | pending | pending   |
| Direct (cross-attn)               | 0.194         | 0.248         | pending     | 0.234         | pending | pending | pending   |
| SHIFTWM (ours)                    | <b>0.180†</b> | <b>0.230†</b> | pending     | <b>0.226†</b> | pending | pending | pending   |
| error reduction vs. best baseline | <b>+7.3%</b>  | <b>+7.1%</b>  | pending     | <b>+3.6%</b>  | pending | pending | pending   |

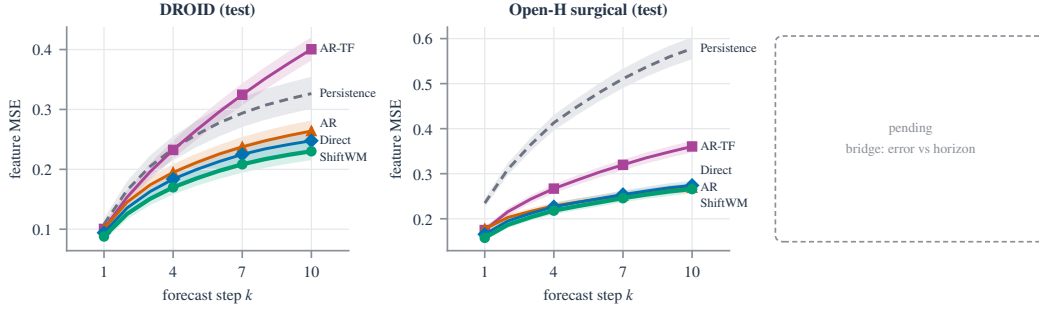


Figure 4: **Forecast error vs. horizon on held-out episodes.** Bands: 95% bootstrap intervals over episodes.

error than Direct, 12.7% lower than the rollout-trained AR and 33.4% lower than the DINO-WM-style AR-TF, whose teacher-forced absolute predictions drift below persistence once rolled out. The same ordering holds on the seven surgical tasks of Open-H, where SHIFTWM is 3.6% below Direct and 4.0% below AR. Figure 4 resolves the error by horizon: recursive predictors accumulate error as they consume their own forecasts (Proposition 3), while SHIFTWM stays lowest at every step.

#### 4.3 SHIFTWM AS A PLUG-IN FOR PUBLISHED WORLD MODELS (Q2)

Because SHIFTWM is an output head, it can be attached to an existing world model. We add it to two strong published models without changing anything else: the head blends the backbone’s prediction  $\mathbf{P}$  with transported observed features,  $\hat{\mathbf{Z}} = (1 - g)\mathbf{P} + g\mathbf{T}$ , with the gate initialised near zero so that training starts from the unmodified model. Each pair shares data, schedule, loss and seed.

**V-JEPA 2-AC.** The released action-conditioned V-JEPA 2 (Assran et al., 2025) (ViT-g encoder, 300M predictor, trained on DROID) removes only 4.1% of persistence error zero-shot on our test windows. Fine-tuning its predictor on our training split raises this to 24.6%; fine-tuning it *with* the SHIFTWM head, under the identical objective and budget, reaches 36.4% (Table 2, top): 11.9 points of skill, 15.7% lower error overall and 16.9% lower on moving patches, for 0.16M added parameters. The learned gate opens to transport on most patches, although the head starts as an exact copy of the backbone.

**DINO-WM.** On DINO-WM’s own benchmarks and code (Zhou et al., 2025), the head reduces the predictor’s latent error by pending% on PushT and pending% on Wall, with better decoded images, and changes planning success from pending% to pending% (PushT) and from pending% to pending% (Wall) under the official planners (Table 2, bottom). Budgets are reduced relative to the original paper and identical for both arms (Appendix C.2).

#### 4.4 WHAT MATTERS (Q3)

Table 3 removes or alters one component at a time on DROID. pending

Table 2: **SHIFTWM as a plug-in head for published world models.** Same data, schedule and loss for each pair; only the head differs. Top: V-JEPA 2-AC on DROID test (its own feature space). Bottom: DINO-WM, official code and protocol.

| V-JEPA 2-AC (ViT-g, 1.3B) on DROID               |  | MSE ↓        | moving ↓     | static ↓     | skill (%) ↑ |
|--------------------------------------------------|--|--------------|--------------|--------------|-------------|
| V-JEPA 2-AC, zero-shot                           |  | 0.504        | 0.815        | 0.400        | 4.1         |
| V-JEPA 2-AC, fine-tuned (2 seeds)                |  | 0.396        | 0.741        | 0.281        | 24.6        |
| V-JEPA 2-AC, fine-tuned + SHIFTWM head (2 seeds) |  | <b>0.334</b> | <b>0.616</b> | <b>0.240</b> | <b>36.4</b> |

| Env   | DINO-WM (ViT-S/14)     | latent err. ↓ | SSIM ↑  | LPIPS ↓ | success (%) ↑ |
|-------|------------------------|---------------|---------|---------|---------------|
| PushT | DINO-WM                | 0.105         | 0.976   | 0.014   | pending       |
| PushT | DINO-WM + SHIFTWM head | pending       | pending | pending | pending       |
| Wall  | DINO-WM                | 0.026         | 0.993   | 0.009   | pending       |
| Wall  | DINO-WM + SHIFTWM head | pending       | pending | pending | pending       |

Table 3: **Ablations on DROID validation.** Each row changes one component of SHIFTWM.

| Variant                                          |                                      | Avg     | $K$     | Rank acc. |
|--------------------------------------------------|--------------------------------------|---------|---------|-----------|
| SHIFTWM (window 7, 3 sources, scaled correction) |                                      | pending | pending | pending   |
| Transport                                        | no transport (= Direct)              | pending | pending | pending   |
|                                                  | global window                        | pending | pending | pending   |
|                                                  | window $w = 3 / w = 11$              | pending | pending | pending   |
| Memory                                           | last frame only ( $S=1$ )            | pending | pending | pending   |
| Correction                                       | tanh-bounded                         | pending | pending | pending   |
|                                                  | none (pure transport)                | pending | pending | pending   |
| Actions                                          | action-free                          | pending | pending | pending   |
|                                                  | + action-contrastive ( $\lambda=1$ ) | pending | pending | pending   |
| Encoder                                          | DINOv2-B/14                          | pending | pending | pending   |
|                                                  | DINOv3-S/16                          | pending | pending | pending   |
|                                                  | DINOv2-S, $4 \times 4$ pooled grid   | pending | pending | pending   |

#### 4.5 CLOSED-LOOP PLANNING (Q3)

All predictors are plugged into the same CEM planner with the official LeWM budget (300 samples  $\times$  30 iterations, 5-block horizon, replanning every 25 steps; Table 8). The released LeWM checkpoint reaches 88% on PushT in our harness (the LeWM paper reports  $96 \pm 2.8$  over training seeds), confirming that the planner is faithful (Appendix C.4). pending

### 5 ANALYSIS

**How does SHIFTWM compose a forecast?** Because Equation (2) is additive, each forecast splits exactly into the kept term  $(1 - g)\mathbf{Z}_0$ , the moved term  $g\mathbf{T}$  and the correction  $\mathbf{r}$  (Figure 2, bottom): the moving arm is carried by transport and the correction concentrates on newly revealed content. Copying measured features also keeps the forecast sharp: at  $k=10$  it retains 0.80 of the true spatial contrast, against 0.77 for Direct and 0.74 for AR (Figure 10).

**Causal role of transport.** We intervene on the trained model (Figure 5). Setting the gate to zero at test time, so that each forecast is  $\mathbf{Z}_0 + \mathbf{r}$ , raises the error by **56.8%** on moving patches and by 35.9% on patches with a high learned gate, but only by 1.8% on static patches: the transport pathway, not the correction, carries the prediction of moving content. Replacing the future actions with those of another episode changes the transport field **2.8 $\times$**  more on moving than on static patches, so the actions steer *where* features are moved. Across all ten deciles of true motion, SHIFTWM has lower error than Direct (**+5.1% to +13.2%**), and the win maps show the gains concentrated on the arm and gripper.

**What does the transport field learn? (Q4)** Converted into an expected source displacement per patch, the field concentrates on the moving arm, gripper and instruments, while the background keeps the identity path (Figure 5); Figure 13 compares it with RAFT optical flow (Teed & Deng, 2020).

**Where do the gains come from?** We split every test window into *moving* patches (the 25% with the largest true feature change, a property of the data shared by all methods) and *static* patches

Table 4: **Planning success (%)** at the official protocol (50 episodes  $\times$  3 seeds). LeWM (end-to-end encoder) is shown for reference.

| Predictor                            | PushT   | TwoRoom | Reacher | s / plan (PushT) |
|--------------------------------------|---------|---------|---------|------------------|
| LeWM (released) (Maes et al., 2026b) | 88.0    | 85.3    | 84.7    | 0.41             |
| AR-TF (DINO-WM-style)                | pending | pending | pending | pending          |
| AR (rollout-trained)                 | pending | pending | pending | pending          |
| Direct                               | pending | pending | pending | pending          |
| SHIFTWM                              | pending | pending | pending | pending          |
| SHIFTWM + action-contrastive         | pending | pending | pending | pending          |

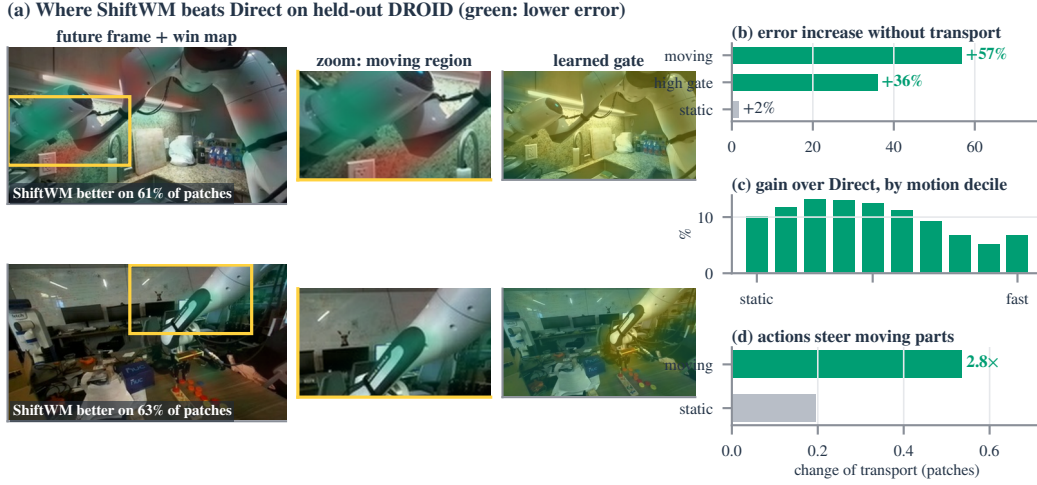


Figure 5: **Where and why SHIFTWM wins (held-out DROID,  $k=10$ ).** (a) Win map (green: SHIFTWM lower error), zoom and gate. (b) Transport knockout. (c) Gain per motion decile. (d) Action steering.

(Table 16, Appendix E.1). Two effects add up. On moving patches, transport predicts where content goes: SHIFTWM has **7.2%** lower error than Direct and **10.0%** lower than AR. On static patches, every other learned predictor is *worse than copying*, since regenerating the grid corrupts background that should not change; SHIFTWM’s identity path keeps it intact, and it is the only learned model below persistence (**2.1%** lower; 15.5% lower than AR).

**Does the moving arm land in the right place?** Read out as a segmentation and compared with SAM 2.1 masks of the true future (Appendix E.2), the arm predicted by SHIFTWM on DROID overlaps the true arm best on moving windows (**+2.0** IoU points over Direct, **+4.2** over AR); on Hamlyn it ties Direct.

**Is the future a transport? A model-free check of Proposition 1.** On moving patches of 3,923 DROID test windows, the single observed feature in the transport window closest to the true future (an *oracle move*) cuts the error of staying put from 0.93 to 0.46 at  $k=10$ : half of the future is already in the observed grid, displaced. SHIFTWM recovers **79%** of this gain (Direct 71%, AR 69%), is better on **64%** of moving patches, and has lower error than Direct and AR in **all 100** horizon-by-motion cells (Figures 11 and 12).

## 6 CONCLUSION

SHIFTWM forecasts by moving what it has already seen: a gated local transport of measured foundation features plus a small learned correction. It is the most accurate predictor on held-out real and surgical robot video, it improves published world models when plugged into them, and its transport field and gate make the predicted motion explicit.

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**Limitations.** SHIFTWM predicts features, not pixels, so pixel metrics rely on a separately trained decoder. Transport can only copy observed content, so newly revealed regions depend on the correction. The surgical benchmark uses scene-camera views of dVRK training tasks, not in-vivo endoscopy.

#### REPRODUCIBILITY STATEMENT

Code, configurations, per-episode evaluation records and trained checkpoints will be released. Appendix B.1 lists every hyperparameter. Dataset splits are deterministic and documented in Appendix C.1. All datasets are public: DROID (CC-BY-4.0), Open-H-Embodiment (CC-BY-4.0), the RLA-WM/IWS release, and the LeWM planning datasets (MIT).

#### ETHICS STATEMENT

The surgical data are public recordings of robotic procedures on phantoms and ex-vivo tissue, released under CC-BY-4.0. We make no clinical claims, and the models are not intended for clinical use.

#### AI USE STATEMENT

In this work, we used generative AI tools to assist with writing and debugging research code, orchestrating experiments, surveying related work, and editing the manuscript text. All AI-assisted code was tested, and every reported number is produced by logged runs that the authors reviewed. Bibliographic entries were verified against the original sources. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

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The appendix follows the order of the main paper.

| Appendix                 | Contents                                                                      |
|--------------------------|-------------------------------------------------------------------------------|
| A Background and setting | the forecasting problem on real episodes, notation, related formulations      |
| B Extended method        | architecture, proofs, plug-in heads                                           |
| C Experimental details   | datasets, baselines, training and compute, planning protocol, statistics      |
| D Extended results       | per-horizon and per-task results, ablations, decoded pixels, probes, planning |
| E Extended analysis      | mechanism analyses, segmentation view, qualitative rollouts, failure cases    |

## A BACKGROUND AND PROBLEM SETTING

### A.1 WHAT THE MODEL SEES AND WHAT IT MUST PREDICT

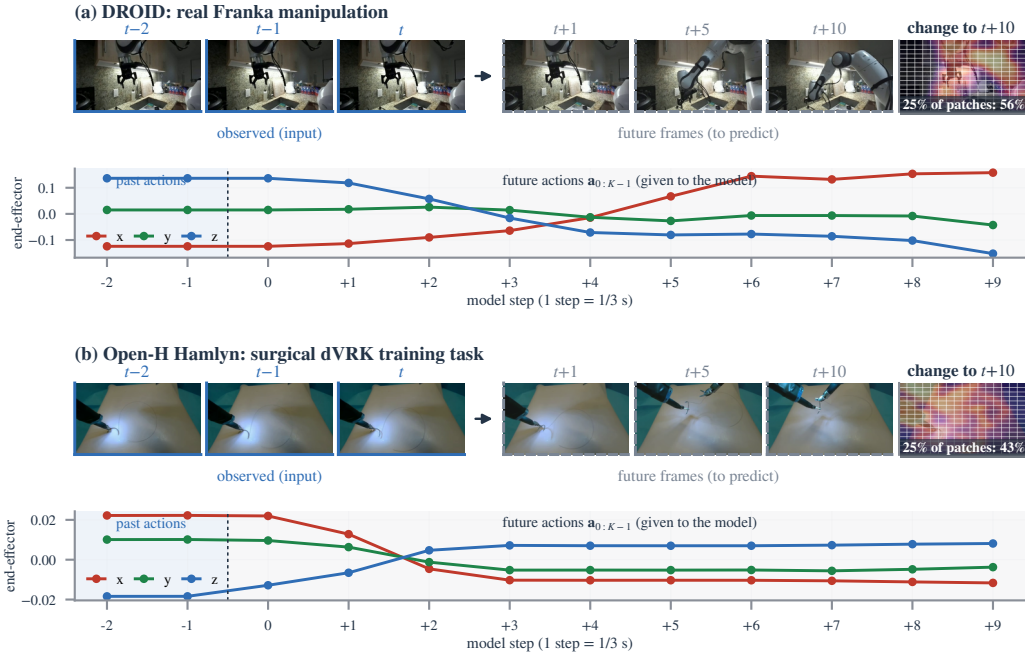


Figure 6: **The forecasting problem on real held-out episodes.** Three observed frames and the future actions are given; features at future steps are predicted. Right: true feature change over the  $16 \times 16$  grid.

Figure 6 shows the task on two held-out episodes, one from DROID (a Franka arm in a real kitchen) and one from the Open-H Hamlyn surgical tasks (a dVRK instrument over a training phantom). The world model receives the last  $H=3$  frames  $t-2, t-1, t$ , encoded by a frozen DINOv2 encoder into  $16 \times 16$  grids of 384-D patch features, the actions applied between them, and a sequence of  $K=10$  future action blocks (the commanded end-effector trajectory, bottom rows). It must predict the feature grids  $\mathbf{Z}_{t+1}, \dots, \mathbf{Z}_{t+10}$ , 3.3 s into the future, without seeing any future frame. The episodes are chosen by a fixed rule (largest feature change among test episodes; Appendix E.3).

The rightmost panels show where the features actually change between  $t$  and  $t+10$ . Change is concentrated on the robot and on the objects it touches: in these two windows a quarter of the patches carry more than 40% of the total change, while large parts of the scene change little. This is the structure that SHIFTWM exploits: most future patch features are already present in the observed grid, displaced by motion, and only a small part is new content.

## A.2 NOTATION

| Symbol                                     | Meaning                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{Z}_t \in \mathbb{R}^{N \times C}$ | frozen patch features of frame $t$ ( $N=256$ patches, $C=384$ for DINOv2-S)                                              |
| $H, K$                                     | number of observed frames (3) and forecast horizon (10 steps of 1/3 s)                                                   |
| $\mathbf{a}_{0:K-1}$                       | future action blocks (commanded poses and gripper states)                                                                |
| $\mathbf{h}_{k,i}$                         | query state of patch $i$ at step $k$ after the action-conditioned decoder                                                |
| $\mathcal{W}(i), w, S$                     | transport window around patch $i$ , its width (7) and number of source frames (3)                                        |
| $\pi_{k,i}(j)$                             | transport weight from observed source patch $j$ to target patch $i$ at step $k$                                          |
| $\mathbf{T}_k, g_k, \mathbf{r}_k$          | transported features, gate and correction; $\hat{\mathbf{Z}}_k = (1 - g_k)\mathbf{Z}_0 + g_k\mathbf{T}_k + \mathbf{r}_k$ |
| $\mathbf{P}_k, \boldsymbol{\varepsilon}_k$ | motion matrix and new content in the generative model of Section 3.1                                                     |
| skill                                      | $1 - \text{MSE}/\text{MSE}_{\text{persistence}}$ , the fraction of persistence error removed                             |

## A.3 RELATED FORMULATIONS

Latent world models such as DINO-WM (Zhou et al., 2025), V-JEPA 2-AC (Assran et al., 2025) and LeWM (Maes et al., 2026b) predict future features of a frozen or jointly trained encoder with a network whose output *replaces* the feature grid: absolute regression, usually rolled out one step at a time. Residual variants predict  $\mathbf{Z}_k - \mathbf{Z}_0$ , which keeps static patches cheap but still requires the network to write the full displacement of every moving patch (Proposition 2). Video prediction in pixel space has long used explicit motion (flow warping, dynamic filters, kernel-based transformations); SHIFTWLM brings this structure to the feature space of foundation models, where features are high-dimensional and semantically stable under motion, so that copying them is accurate. Unlike optical-flow pipelines, the transport is never supervised with flow: it is learned end-to-end from the feature forecasting loss, conditioned on future actions, and is used at every horizon directly from the observed grid.

## B EXTENDED METHOD

### B.1 ARCHITECTURE

**Architecture.** Width  $d=384$ , 6 heads, MLP ratio 4, memory encoder  $L_e=4$  layers of full self-attention over the  $H \cdot N=768$  tokens of  $H=3$  observed  $16 \times 16$  grids. Query decoder:  $L_d=4$  layers, each with self-attention over the  $N=256$  query tokens, cross-attention to memory and an MLP, modulated by AdaLN-zero on  $\mathbf{c}_k$ . Action embedding: a 2-layer MLP followed by a single-layer unidirectional GRU of width  $d$ . Transport: key dimension 64, window  $w=7$  over  $S=3$  source frames (147 candidates per query), learned identity bias initialised at  $\beta=4$ , and gate bias  $-3$ . Correction head: LayerNorm, then a zero-initialised linear map to  $C$ , then a per-channel scale initialised at 1. Parameters: 21.75M (SHIFTWLM) and 21.70M (Direct, AR, AR-TF).

### B.2 PROOFS

**Proof of Proposition 1.** The conditional mean minimises expected squared error. Since  $\mathbf{Z}_0$  is part of the context and  $\mathbb{E}[\boldsymbol{\varepsilon}_k \mid \mathbf{P}_k, \mathbf{Z}_0, \mathbf{a}] = \mathbf{0}$ ,  $\mathbb{E}[\mathbf{Z}_k \mid c] = \mathbb{E}[\mathbf{P}_k \mid c] \mathbf{Z}_0 + \mathbb{E}[\mathbb{E}[\boldsymbol{\varepsilon}_k \mid \mathbf{P}_k, c] \mid c] = \bar{\mathbf{P}}_k \mathbf{Z}_0$ . Each row of  $\mathbf{P}_k$  is a one-hot vector, so each row of  $\bar{\mathbf{P}}_k$  is a probability vector supported on the same indices, hence on  $\mathcal{W}(i)$  for row  $i$ . For  $g = 1$  and  $\mathbf{r} = \mathbf{0}$ , Equation (2) gives  $\hat{\mathbf{Z}}_{k,i} = \sum_{j \in \mathcal{W}(i)} \pi_{k,i}(j) \mathbf{Z}_{0,j}$ . Softmax weights with logits  $\ell_j = \log \bar{P}_{k,i,j}$  on the support and  $\ell_j \rightarrow -\infty$  elsewhere converge to row  $i$  of  $\bar{\mathbf{P}}_k$ , so  $\hat{\mathbf{Z}}_k \rightarrow \bar{\mathbf{P}}_k \mathbf{Z}_0$ . With  $S > 1$  source frames the same argument applies to a motion model whose rows select patches of any observed frame.  $\square$

**Proof of Proposition 2.** An additive head outputs  $\mathbf{Z}_0 + \mathbf{R}$ ; matching  $\mathbf{P}\mathbf{Z}_0$  requires  $\mathbf{R} = (\mathbf{P} - \mathbf{I})\mathbf{Z}_0$ , whose  $i$ -th row is  $\mathbf{z}_{0,\pi(i)} - \mathbf{z}_{0,i}$ ; summing squared row norms gives the claim. For SHIFTWLM, the one-hot transport onto  $\pi(i)$  (the limit above) with  $g = 1$  reproduces  $\mathbf{P}\mathbf{Z}_0$  with  $\mathbf{r} = \mathbf{0}$ .  $\square$

**Proof of Proposition 3.** Let  $e_j = \|\hat{\mathbf{Z}}_j - \mathbf{Z}_j\|$  for the recursive rollout  $\hat{\mathbf{Z}}_j = f(\hat{\mathbf{Z}}_{j-1})$  and  $\mathbf{Z}_j = f(\mathbf{Z}_{j-1}) + \boldsymbol{\eta}_j$  with  $\|\boldsymbol{\eta}_j\| \leq \delta$ . Then  $e_j \leq \|f(\hat{\mathbf{Z}}_{j-1}) - f(\mathbf{Z}_{j-1})\| + \|\boldsymbol{\eta}_j\| \leq L e_{j-1} + \delta$  with  $e_0 = 0$ , so  $e_k \leq \delta \sum_{j=0}^{k-1} L^j$ . A direct predictor computes  $\hat{\mathbf{Z}}_k$  from observed inputs only, so its error at horizon  $k$  does not depend on its predictions at other horizons.  $\square$

### B.3 PLUG-IN HEADS FOR PUBLISHED WORLD MODELS

For a published backbone with prediction  $\mathbf{P}_k$ , the head outputs  $\hat{\mathbf{Z}}_k = (1 - g_k)\mathbf{P}_k + g_k\mathbf{T}_k$ , where  $\mathbf{T}_k$  is the local  $7 \times 7$  soft transport (with identity bias) of *observed* encoder tokens, and the gate  $g_k = \sigma(\mathbf{w}^\top \mathbf{h} + b)$  has  $\mathbf{w} = \mathbf{0}$ ,  $b = -4$  at initialisation, so training starts numerically from the unmodified backbone. Nothing else in the backbone, loss or schedule changes. **V-JEPA 2-AC**. Queries come from the predictor’s normalised hidden state for each output block (1024-D), keys and values from the layer-normalised ViT-g/16 tokens ( $16 \times 16$ , 1408-D) of the last  $S=3$  measured frames, with a learned per-source key offset (0.16M parameters). The fine-tuning objective is the released one (teacher-forced L1 plus rollout L1). During rollouts the predictor consumes  $\hat{\mathbf{Z}}_k$  as its own prediction, while  $\mathbf{T}_k$  only ever reads measured tokens. **DINO-WM**. We subclass the official ViTPredictor (commit 0a9492f, otherwise unmodified): queries come from the predictor’s output token, keys and values from its input token at the same slot, on the  $14 \times 14$  DINOv2 patch grid ( $S=1$ ); proprioceptive and action channels are left untouched. The extra parameters are drawn from a private random generator, so data order, initialisation and dropout of the base predictor are identical in both arms.

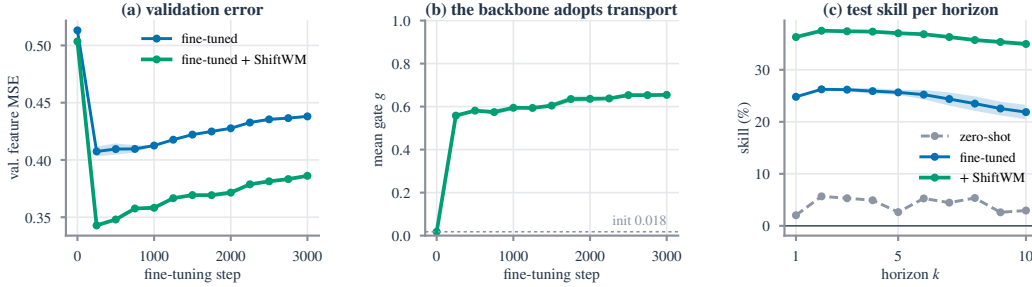


Figure 7: **V-JEPA 2-AC with and without the SHIFTWm head (DROID)**. (a) Validation error while fine-tuning (2 seeds). (b) Mean gate. (c) Test skill per horizon.

**Plug-in dynamics.** Figure 7 shows both fine-tuning arms over their full 3,000-step budget. With the head, validation error is lower at every evaluation step for both seeds. The gate starts at 0.018, so the model begins as the released predictor, and opens to about 0.55 within the first 250 steps, rising to about 0.65: the backbone learns to route most of its forecast through transport of measured tokens. Both arms reach their best validation error early and then slowly overfit our training split; the test numbers use the best validation checkpoint of each run. The skill gain holds at every horizon and grows from +11.5 points at  $k=1$  to +13.1 at  $k=10$ .

## C EXPERIMENTAL DETAILS

### C.1 DATASETS AND SPLITS

Table 5: **Datasets**. Step = one model time step. All splits are disjoint at the unit shown.

| Dataset                   | Domain              | Episodes (train/val/test) | Step        | Action dim | Split unit       | License   |
|---------------------------|---------------------|---------------------------|-------------|------------|------------------|-----------|
| DROID (24 shards)         | real Franka, 3 cams | 851 / 143 / 132           | 0.33 s      | 35         | session          | CC-BY-4.0 |
| Open-H Hamlyn (7 tasks)   | surgical dVRK       | 708 / 152 / 153           | 0.33 s      | 80         | episode          | CC-BY-4.0 |
| IWS PushT / Box / Rope    | real manipulation   | pending                   | pending     | pending    | official handles | pending   |
| BridgeData V2 (subset)    | real WidowX         | 518 / 57 / 159            | 0.4 s       | 14         | official split   | CC-BY-4.0 |
| RT-1 Fractal (subset)     | real mobile robot   | 8,657 / 942 / 1,101       | 0.33 s      | 7          | episode          | CC-BY-4.0 |
| PushT / TwoRoom / Reacher | simulation          | pending                   | 5 env steps | 10         | episode          | MIT       |

**DROID.** We use 24 official full-release shards (22.0 GB) containing 1,126 successful episodes. Splits are assigned by recording session with a salted hash, so no session appears in two splits. Frames are subsampled every 5 native steps (15 Hz  $\rightarrow$  3 Hz), and each action block concatenates the 5 commanded Cartesian poses and gripper positions ( $5 \times 7=35$ ).

**Open-H-Embodiment Hamlyn.** Seven dVRK training tasks, released under CC-BY-4.0 in LeRobot format. We use the `observation.images.color` stream, which the release labels “endoscope” but which is an Intel RealSense scene camera (848×480). One step is 1/3 s: stride 10 at 30 Hz, or stride 5 for the 15 Hz tissue-lifting task. Each action block holds five commands at 15 Hz, each an 8-D pose-and-gripper target for each arm ( $5 \times 16=80$ ). Episodes shorter than 14 steps (6 of 1,019) are removed. Splits are 70/15/15 per task, assigned by a hash of the episode id.

**IWS.** pending

**Planning suites.** PushT, TwoRoom and Reacher expert datasets from the LeWM release (HDF5). Episodes are split disjointly into train and validation sets for world-model training. Evaluation follows the official protocol (Appendix C.4).

## C.2 BASELINES

**Matched predictors.** All learned baselines share SHIFTWm’s memory encoder, decoder, width, depth, optimiser and schedule, and differ only in the output head and rollout. This isolates the contribution of transport from those of capacity and spatial communication. Direct has the same cross-patch attention as SHIFTWm and differs only in lacking the transport and gate.

**DINO-WM-style (AR-TF).** Following Zhou et al. (2025), the predictor regresses absolute next-step patch features with teacher forcing and is rolled out autoregressively at test time. We use the same frozen DINOv2 patches at native resolution.

**External reference: V-JEPA 2-AC.** We evaluate the released V-JEPA 2-AC (Assran et al., 2025) (ViT-g encoder and 300M action-conditioned predictor, trained on DROID) zero-shot on exactly our test windows, with its own preprocessing, action convention and autoregressive rollout. Because feature spaces differ, we compare the fraction of persistence error each model removes in its own space (Table 6); we also fine-tune it on our training split with and without the SHIFTWm head under an identical budget. Its training data likely include our test trajectories.

Table 6: **External models on the same DROID test windows:** % of persistence error removed (skill  $\uparrow$ ), each in its own feature space.

| Model                                           | Feature space  | Skill (%) [95% CI] |
|-------------------------------------------------|----------------|--------------------|
| V-JEPA 2-AC (zero-shot, ViT-g, 1.3B)            | V-JEPA 2 ViT-g | 4.1 [0.6, 7.3]     |
| V-JEPA 2-AC fine-tuned on our split (2 seeds)   | V-JEPA 2 ViT-g | 24.6               |
| V-JEPA 2-AC fine-tuned + SHIFTWm head (2 seeds) | V-JEPA 2 ViT-g | <b>36.4</b>        |
| AR-TF (DINO-WM-style)                           | DINOv2-S       | -8.2               |
| AR                                              | DINOv2-S       | 17.5               |
| Direct                                          | DINOv2-S       | 22.3               |
| SHIFTWm (ours)                                  | DINOv2-S       | <b>28.0</b>        |

**Earlier variant.** Our earlier compact variant (4×4 pooled grids, per-patch temporal predictor, tanh-bounded correction) is reported for continuity in Appendix D.2.

## C.3 TRAINING AND COMPUTE

**Training.** AdamW ( $\beta=(0.9, 0.95)$ ), weight decay 0.05), peak learning rate  $3 \times 10^{-4}$  with 800 warm-up steps and cosine decay, 16k steps, batch 48, bf16 autocast, gradient clipping at 1.0. Validation MSE (averaged over horizons, windows at stride 2) is computed every 2k steps, and the best checkpoint is used for testing. AR is trained through its full 10-step rollout. AR-TF is trained with teacher forcing and rolled out at test time. All features for a split are kept in GPU memory as float16 and windows are gathered by index.

**Compute.** All experiments ran on NVIDIA H200 GPUs, packing at most two learned predictors per GPU. pending

Table 7: **Hyperparameters** (shared by all learned predictors unless noted).

|                                       |                       |                     |                                |
|---------------------------------------|-----------------------|---------------------|--------------------------------|
| Width / heads / MLP ratio             | 384 / 6 / 4           | Optimiser           | AdamW, $\beta=(0.9, 0.95)$     |
| Memory encoder layers $L_e$           | 4                     | Peak LR / warm-up   | $3 \times 10^{-4}$ / 800 steps |
| Query decoder layers $L_d$            | 4                     | Schedule            | cosine, 16k steps              |
| History $H$ / horizon $K$             | 3 / 10                | Batch size          | 48 windows                     |
| Transport window $w$ / sources $S$    | 7 / 3                 | Weight decay / clip | 0.05 / 1.0                     |
| Key dim / identity bias init          | 64 / 4.0              | Precision           | bf16 autocast                  |
| Gate bias init                        | -3                    | Selection           | best val. MSE (every 2k steps) |
| Action-contrastive $\lambda$ / margin | 1.0 / 0.05 (planning) | Seeds               | 0, 1, 2                        |

Table 8: **Official planning protocol** (le-wm@8edfeb33, stable-worldmodel@4821c8e6).

|                      |                                                                                          |
|----------------------|------------------------------------------------------------------------------------------|
| Planner              | CEM, 300 samples $\times$ 30 iterations, 30 elites, initial variance 1, warm start       |
| Horizon / replanning | 5 action blocks of 5 env steps; replan every 25 steps                                    |
| Episodes / seeds     | 50 episodes per planner seed; seeds 42, 43, 44                                           |
| Goal                 | dataset frame 25 steps later in the same trajectory                                      |
| Budget               | 50 environment steps                                                                     |
| Cost                 | squared distance between final predicted latent and goal latent                          |
| Success              | PushT: position $< 20$ px and angle $< \pi/9$ ; TwoRoom: $< 16$ px; Reacher: joint match |

#### C.4 PLANNING PROTOCOL

#### C.5 STATISTICAL PROCEDURES

Each test episode contributes the mean of its windows (stride 2). For each method and seed we average over episodes, and tables report the mean over three training seeds. For a paired comparison between methods A and B, we compute per-episode differences (averaged over seeds) and draw 10,000 bootstrap resamples of episodes to form a percentile 95% interval. For DROID, we resample sessions instead of episodes to respect the session structure. Primary contrasts (SHIFTWM against each baseline on each dataset) are Holm-adjusted within each dataset.

### D EXTENDED RESULTS

#### D.1 FULL FORECASTING RESULTS

Table 9: **Per-horizon test MSE on DROID** (mean over 3 seeds).

| Method                | $k=1$        | 2            | 3            | 4            | 5            | 6            | 7            | 8            | 9            | 10           |
|-----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Persistence           | 0.108        | 0.165        | 0.205        | 0.235        | 0.258        | 0.278        | 0.294        | 0.307        | 0.317        | 0.326        |
| AR-TF (DINO-WM-style) | 0.101        | 0.153        | 0.196        | 0.232        | 0.265        | 0.296        | 0.325        | 0.352        | 0.377        | 0.400        |
| AR (rollout-trained)  | 0.101        | 0.145        | 0.174        | 0.195        | 0.211        | 0.225        | 0.237        | 0.248        | 0.256        | 0.264        |
| Direct (cross-attn)   | 0.094        | 0.136        | 0.163        | 0.184        | 0.199        | 0.213        | 0.225        | 0.234        | 0.241        | 0.248        |
| SHIFTWM (ours)        | <b>0.088</b> | <b>0.126</b> | <b>0.151</b> | <b>0.170</b> | <b>0.185</b> | <b>0.198</b> | <b>0.208</b> | <b>0.217</b> | <b>0.224</b> | <b>0.230</b> |

pending

#### D.2 ADDITIONAL ABLATIONS

**Training recipes.** Because all learned predictors reach their best validation error early (Appendix B.1), we tested whether stronger regularisation or more data helps (Table 12): training on both exterior cameras with EMA weights, dropout and memory-token dropout, and adding explicit correlation (cost-volume) features between consecutive observed grids to every method. pending pending

#### D.3 DECODED-PIXEL METRICS

For each dataset we train one feature-to-RGB decoder on true training features (L1 + LPIPS) and apply it unchanged to every method’s forecasts, so differences reflect the predicted features only.

pending

Table 10: **Open-H Hamlyn test MSE per surgical task** (mean over horizons and 3 seeds).

| Method                | Knot         | Needle       | Peg          | Sut. 1       | Sut. 2       | Lift         | Retract      |
|-----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Persistence           | 0.600        | 0.522        | 0.350        | 0.376        | 0.471        | 0.546        | 0.575        |
| AR-TF (DINO-WM-style) | 0.401        | 0.353        | 0.246        | 0.223        | 0.297        | 0.335        | 0.314        |
| AR (rollout-trained)  | 0.329        | 0.304        | 0.201        | 0.187        | 0.248        | 0.254        | 0.247        |
| Direct (cross-attn)   | 0.322        | 0.304        | 0.201        | 0.184        | 0.244        | 0.257        | 0.253        |
| SHIFTWM (ours)        | <b>0.312</b> | <b>0.297</b> | <b>0.194</b> | <b>0.177</b> | <b>0.237</b> | <b>0.241</b> | <b>0.239</b> |

Table 11: **IWS per task** on the official RLA-WM handles.

| Method      | PushT   |         | Box     |         | Rope    |         |
|-------------|---------|---------|---------|---------|---------|---------|
|             | MSE     | LPIPS   | MSE     | LPIPS   | MSE     | LPIPS   |
| Persistence | pending | pending | pending | pending | pending | pending |
| AR          | pending | pending | pending | pending | pending | pending |
| Direct      | pending | pending | pending | pending | pending | pending |
| SHIFTWM     | pending | pending | pending | pending | pending | pending |

#### D.4 PROBING PREDICTED FEATURES

Ridge probes map mean- and  $4 \times 4$ -pooled features to the commanded DROID end-effector position or the measured Hamlyn instrument-tip positions (both arms pooled). They are fit on true training features, with the ridge strength chosen on validation, and are evaluated on true versus predicted test features at  $k=10$ . pending

#### D.5 PLANNING RESULTS AND FAITHFULNESS OF THE HARNESS

**Faithfulness check.** With the released LeWM PushT checkpoint and the official protocol, our harness reaches 45/50, 44/50 and 43/50 successes for planner seeds 42/43/44 (mean 88%). An adapter check confirms that our pluggable predictor interface produces exactly the native LeWM planning costs (maximum absolute difference 0). The LeWM paper reports  $96 \pm 2.8$ , averaged over training seeds, whereas the released checkpoint is a single training seed. On TwoRoom and Reacher the released checkpoints reach 43/41/44 and 42/44/41 successes (85.3% and 84.7%), against 87 and 86 reported by [Maes et al. \(2026b\)](#). Planning takes 0.41 s (PushT) and about 1.5 s (TwoRoom, Reacher) per CEM plan on one H200.

#### D.6 ACTION SENSITIVITY AND EFFICIENCY

**Accuracy versus action sensitivity.** A forecast that stays close to  $\mathbf{Z}_0$  can score well on average while barely responding to the commanded actions, and a planner then cannot rank candidate action sequences. We measure this directly as the rate at which true future actions yield lower error than actions from another episode (Figure 9, left). Anchored predictors can reach low open-loop error while reacting weakly to actions; the action-contrastive objective moves SHIFTWM toward high accuracy and high sensitivity. pending

**Efficiency.** Table 15 compares parameters, compute per forecast step and latency. Transport adds pending parameters. Because SHIFTWM and Direct decode all  $K$  steps in parallel, a  $K$ -step forecast needs one encoder pass, whereas recursive predictors need  $K$  (pending).

### E EXTENDED ANALYSIS AND QUALITATIVE RESULTS

#### E.1 ANALYSIS DETAILS

**Feature-space geometry.** Figure 11 is the data counterpart of Figure 3. Panel (a) shows, for one patch at a time, the  $Sw^2=147$  observed candidate features of its transport window projected to 2-D by PCA, sized by the learned transport weight, together with the observed feature at the same location, the true future and the three forecasts. The patches are chosen by a fixed rule: in the test windows at the 25th, 50th and 75th percentile of mean true change, the moving, high-gate ( $g>0.5$ )

Table 12: **Training recipes and add-ons** evaluated on DROID (test MSE, seed 0). All methods receive each change.

| Recipe                     | AR-TF   | AR           | Direct | SHIFTWM      |
|----------------------------|---------|--------------|--------|--------------|
| base (16k steps)           | 0.271   | 0.203        | 0.193  | <b>0.179</b> |
| base, short (8k steps)     | pending | pending      | 0.195  | <b>0.180</b> |
| + 2nd camera, EMA, dropout | pending | <b>0.209</b> | 0.201  | <b>0.181</b> |
| + correlation features     | pending | pending      | 0.203  | <b>0.195</b> |

**Pending results.** Validation MSE vs. transport window size  $w \in \{3, 5, 7, 11, \text{global}\}$  and number of source frames  $S \in \{1, 3\}$ .

Figure 8: **Transport window and memory size.**

patch with the median change. They are illustrations only; individual patches can favour any method (patch 1 favours AR). Panels (b) and (c) aggregate over all moving patches of 3,923 windows from 130 test episodes. The oracle move picks, for each patch, the single observed candidate closest to the true future; it is a model-free reference, not an achievable predictor, since it uses the future to choose. Gain recovered is  $(\text{err}_{\text{stay}} - \text{err}_{\text{model}})/(\text{err}_{\text{stay}} - \text{err}_{\text{oracle}})$ . Paired episode-bootstrap 95% intervals of the moving-patch error difference at  $k=10$  exclude zero against both Direct and AR (results/v2/analysis/geometry/summary.json).

**Anatomy of the gains.** Figure 12 uses all 3,923 test windows. In (a,b) every patch at every horizon is assigned to a decile of its true change  $\|\mathbf{z}_{k,i} - \mathbf{z}_{0,i}\|^2$  (deciles pooled over horizons), and each cell reports  $1 - \text{err}_{\text{SHIFTWM}}/\text{err}_{\text{baseline}}$ ; the  $10 \times 10$  grid is drawn as filled contours after cubic interpolation, with the maximum marked. Against Direct the gain is largest for moderate-to-large motion at long horizons; against AR it is largest on nearly static patches, where the recursive rollout accumulates error on content that should not change. (c) The forecast departs from  $\mathbf{Z}_0$  as  $g(\mathbf{T} - \mathbf{Z}_0) + \mathbf{r}$  exactly; on moving patches the mean gate rises from 0.51 at  $k=1$  to 0.85 at  $k=10$ , and the moved term carries 87% of the squared departure on average, so the correction remains a small residual. (d) On moving patches SHIFTWM recovers a larger share of the oracle-move gain than both baselines at every horizon, reaching 79% at  $k=10$  against 71% (Direct) and 69% (AR).

pending

## E.2 SEGMENTATION VIEW OF THE FORECASTS

**Does the moving arm land in the right place?** Feature error does not show *what* moves, so we read a segmentation out of every forecast (Figure 14). SAM 2.1 (Ravi et al., 2025) tracks the robot arm (the instruments on Hamlyn) through the true future frames, and each forecast patch is labelled foreground or background by nearest-neighbour transfer from the observed frame alone (protocol and quality control in Appendix E.2). On 1,332 held-out DROID windows the arm read out of SHIFTWM overlaps the true arm best: averaged over  $k$ , IoU 0.65 vs. 0.64 (Direct), 0.63 (AR) and 0.62 (persistence). On the half of windows where the arm moves most, the paired gain is **+2.0** IoU points over Direct (95% CI [+1.2, +2.7]) and **+4.2** over AR ([+3.3, +5.1]), and it grows with the horizon (+2.7 over Direct at  $k=10$ ). On Hamlyn, SHIFTWM and Direct place the instruments equally well (+0.1, CI [-0.2, +0.4]) and SHIFTWM is only marginally ahead of AR (+0.4, CI [+0.1, +0.8]); every learned model is far ahead of persistence. Single windows vary (Figure 14a); the claim rests on the averages.

**Protocol.** We ask whether the moving part of the scene lands where it should in each forecast, without decoding pixels. *Reference masks.* On the observed frame  $t$  (the last history frame) of

Table 13: **Decoded-pixel quality at  $k=10$** . One decoder per dataset for all methods; the first row decodes true features.

| Method                                 | DROID           |                 |                    | Open-H Hamlyn   |                 |                    |
|----------------------------------------|-----------------|-----------------|--------------------|-----------------|-----------------|--------------------|
|                                        | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ |
| Decoder on true features (upper bound) | pending         | pending         | pending            | pending         | pending         | pending            |
| Persistence                            | pending         | pending         | pending            | pending         | pending         | pending            |
| AR-TF (DINO-WM-style)                  | pending         | pending         | pending            | pending         | pending         | pending            |
| AR                                     | pending         | pending         | pending            | pending         | pending         | pending            |
| Direct                                 | pending         | pending         | pending            | pending         | pending         | pending            |
| SHIFTWM                                | pending         | pending         | pending            | pending         | pending         | pending            |

Table 14: **Robot state decoded from predicted features at  $k=10$**  by linear probes (DROID: end-effector, cm; Hamlyn: instrument tips, mm).

| Method                             | DROID EE position |                        | Hamlyn tip position |                        |
|------------------------------------|-------------------|------------------------|---------------------|------------------------|
|                                    | MAE $\downarrow$  | Pearson $r$ $\uparrow$ | MAE $\downarrow$    | Pearson $r$ $\uparrow$ |
| True future features (upper bound) | pending           | pending                | pending             | pending                |
| Persistence                        | pending           | pending                | pending             | pending                |
| AR-TF (DINO-WM-style)              | pending           | pending                | pending             | pending                |
| AR                                 | pending           | pending                | pending             | pending                |
| Direct                             | pending           | pending                | pending             | pending                |
| SHIFTWM                            | pending           | pending                | pending             | pending                |

every held-out window (test split; the planning suites have no test split, so their validation split is used; stride 2 over all episodes, and an evenly spaced subset of 1,500 windows where there are more), Grounding DINO (Liu et al., 2024) detects a fixed per-dataset text prompt (DROID, Bridge, RT-1: “a robot arm”; Hamlyn: “a surgical instrument”, up to two non-overlapping boxes; IWS: the manipulated T block, box or rope). Its top box prompts SAM 2.1 (Hiera-L) (Ravi et al., 2025), which segments frame  $t$  and *tracks* the mask through the true future frames  $t+1, \dots, t+K$  with its video memory. Frames are segmented at their native aspect ratio; each mask keeps its largest connected component and any component with  $\geq 25\%$  of its area. For the rendered planning suites a text prompt is unreliable, so an exact colour threshold of the agent/object is used instead. *Quality control*. A window is kept only if the detection score is  $\geq 0.3$ , the mask covers 3–128 of the 256 patches at  $t$ , the track is never lost and the mask area never changes by more than  $3\times$  between consecutive frames; on DROID this keeps 1,332 of 1,500 windows (rejected: 104 without a confident detection, 45 with an arm too small or too large at  $t$ , 8 lost tracks, 11 area jumps), and on Hamlyn 1,017 of 1,500. *Reading a mask out of a forecast*. Masks are area-averaged onto the  $16\times 16$  patch grid (a patch is foreground if  $\geq 50\%$  covered). Each patch of a forecast  $\hat{\mathbf{Z}}_k$  is labelled by nearest-neighbour transfer from the observed frame only: it is foreground if its most similar (cosine) foreground patch of  $\mathbf{Z}_0$  is more similar than its most similar background patch. This labeller was fixed in a DROID pilot as the one with the highest IoU when applied to the *true* future features, a criterion that involves no forecast (on DROID at  $k=10$ : nearest-neighbour 0.70,  $k$ -means prototypes 0.60, single prototypes 0.42). *Metric*. IoU between the forecast mask and the tracked SAM mask of the true future frame, per horizon and averaged over  $k$ , on all windows and on *moving* windows (overlap of the masks at  $t$  and  $t+K$  below the median). CIs are 95% bootstrap intervals over episodes; gains are paired differences per episode. *Examples* are chosen by a fixed rule that ignores the compared methods: the QC-passing windows at the 90th and 50th percentile of motion. Checkpoints are the final ones used in the main tables.

**Results.** Table 17 lists IoU per dataset and method (Figures 15 and 16 show both example windows per dataset). On DROID, SHIFTWM has the highest IoU at every reported horizon, on all and on moving windows, with paired CIs that exclude zero against both Direct and AR. On Hamlyn the instruments move less within ten steps and all three learned predictors are within half an IoU point: SHIFTWM ties Direct and is marginally ahead of AR; the segmentation view does not separate the methods there. Datasets whose ShiftWM/Direct checkpoints are not trained yet are marked pending and are filled in by re-running `scripts/v2/segments_all.sh`.

**Caveats.** (i) The readout measures where a forecast puts content that *looks like* the observed foreground; it cannot credit content that was not visible at  $t$ . (ii) The labeller is imperfect: applied to the true future features it reaches IoU 0.70 on DROID at  $k=10$ , so absolute values are lower bounds on localisation quality; on Hamlyn several forecasts slightly exceed this reference, because smoother

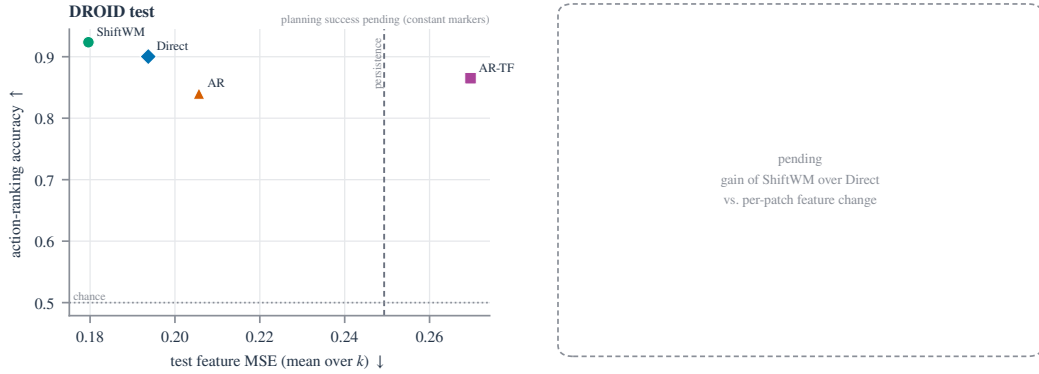


Figure 9: **Left:** forecast error vs. action-ranking accuracy (marker size: planning success). **Right:** gain of SHIFTFM over Direct by amount of motion.

Table 15: **Efficiency** of a 10-step forecast on one H200 (batch 1, bf16).

| Predictor  | Params (M) | GFLOPs / 10 steps | Latency (ms) | Train GPU-h |
|------------|------------|-------------------|--------------|-------------|
| AR-TF / AR | 21.70      | pending           | pending      | pending     |
| Direct     | 21.70      | pending           | pending      | pending     |
| SHIFTFM    | 21.75      | pending           | pending      | pending     |

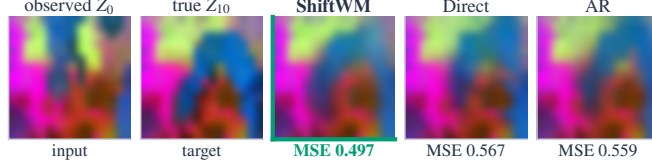
forecasts can be labelled more easily than the true features. (iii) SAM 2.1 occasionally includes a grasped object or tool-attached structure in the tracked mask (Figure 16, bottom row); QC removes lost or exploding tracks but not such merges. On Hamlyn most rejected windows (363 of 483) are close-ups in which the instruments cover more than half of the frame. (iv) One seed per method.

### E.3 QUALITATIVE RESULTS

Table 16: **Where the gains come from** (DROID test, feature MSE  $\downarrow$  by region; skill = error removed vs. persistence). [Interim: seeds completed – DROID: ar 2/3, ar-tf 2/3, direct 2/3, shiftwm 2/3; Hamlyn: ar 0/3, ar-tf 0/3, direct 0/3, shiftwm 0/3.]

| Method                | moving patches | static patches | all          | skill $\uparrow$ |
|-----------------------|----------------|----------------|--------------|------------------|
| Persistence           | 0.726          | 0.090          | 0.249        | 0.0%             |
| AR-TF (DINO-WM-style) | 0.592          | 0.161          | 0.268        | -7.6%            |
| AR (rollout-trained)  | 0.499          | 0.105          | 0.203        | 18.4%            |
| Direct (cross-attn)   | 0.484          | 0.097          | 0.194        | 22.3%            |
| SHIFTWM (ours)        | <b>0.449</b>   | <b>0.088</b>   | <b>0.179</b> | <b>28.3%</b>     |

(a) Forecasts in feature space (PCA  $\rightarrow$  RGB,  $k=10$ )



(b) Forecasts stay sharp

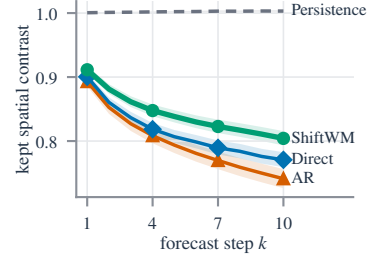


Figure 10: **Forecasts in feature space.** (a) Held-out forecasts at  $k=10$  (shared PCA to colour). (b) Spatial contrast kept by each forecast, relative to the truth (95% CIs over test episodes).

(a) One patch in feature space (PCA of its observed candidates; size = transport weight)

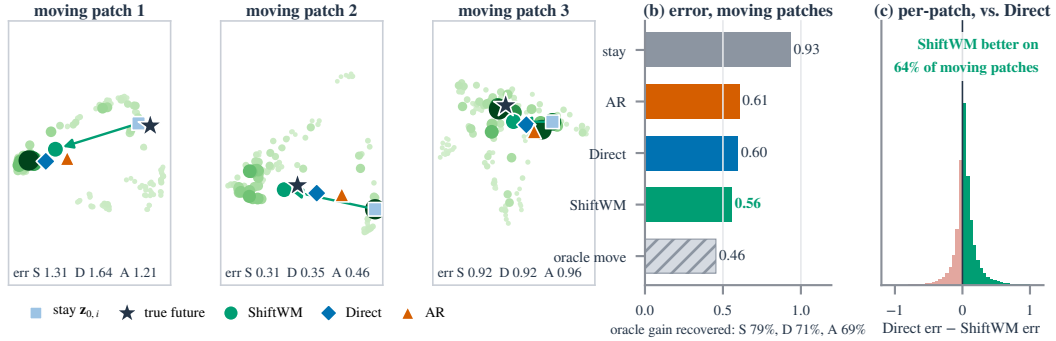


Figure 11: **Feature-space geometry on held-out DROID** ( $k=10$ ). (a) Three moving patches (fixed selection rule) in a 2-D PCA of their observed candidates. (b) Moving-patch error, with the model-free oracle move. (c) Per-patch error difference to Direct.

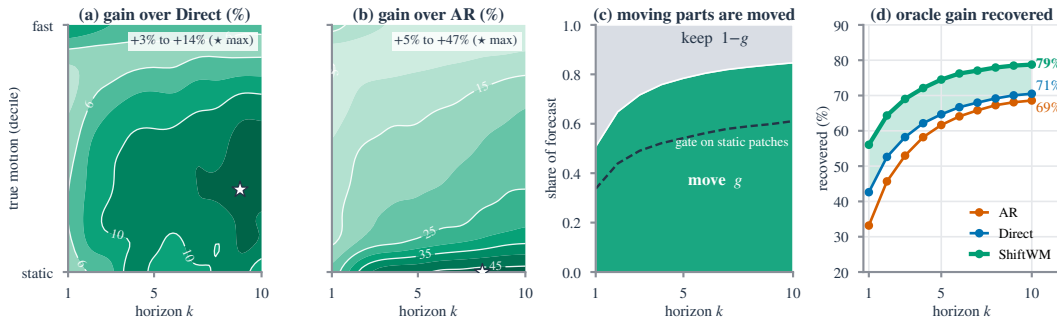


Figure 12: **Anatomy of the gains (held-out DROID).** (a,b) Error reduction over horizon and motion decile. (c) Gate on moving (area) and static (dashed) patches. (d) Share of the oracle-move gain recovered on moving patches.



Figure 13: Transport vs. optical flow.

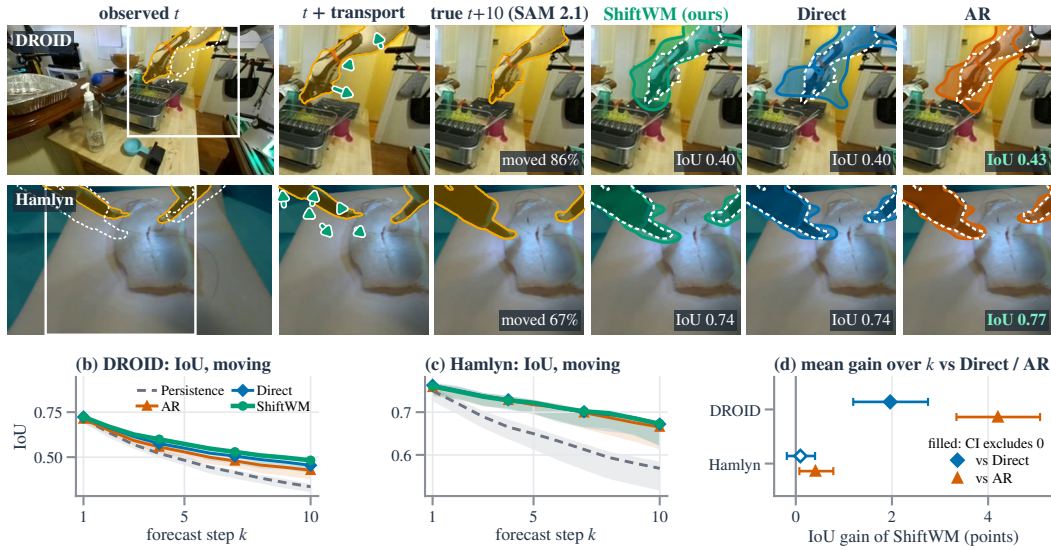


Figure 14: **Where does the arm go?** (a) Held-out windows at the 90th percentile of motion (fixed rule): SAM 2.1 mask at  $t$  and  $t+10$  (yellow; dashed white: true future), SHIFW transport arrows, and the region each  $k=10$  forecast labels as arm/instrument (heat: soft probability; IoU in the corner). (b,c) IoU vs. horizon on moving windows (95% CIs over episodes). (d) Paired gain of SHIFW, averaged over  $k$ ; filled: CI excludes 0.

Table 17: **Segmentation view: IoU of the foreground read out of each forecast** (horizon-averaged,  $k=5$  and  $k=K$ ; all and moving windows). **Green**: paired 95% CI of SHIFTWM minus *both* Direct and AR excludes zero.

| Dataset                                                                                                     | Method      | all windows   |              |              | moving windows |              |
|-------------------------------------------------------------------------------------------------------------|-------------|---------------|--------------|--------------|----------------|--------------|
|                                                                                                             |             | avg. over $k$ | $k=5$        | $k=K$        | avg. over $k$  | $k=K$        |
| DROID                                                                                                       | Persistence | 0.617         | 0.612        | 0.536        | 0.486          | 0.335        |
|                                                                                                             | AR          | 0.628         | 0.624        | 0.551        | 0.535          | 0.427        |
|                                                                                                             | Direct      | 0.644         | 0.638        | 0.576        | 0.557          | 0.454        |
|                                                                                                             | SHIFTWM     | <b>0.654</b>  | <b>0.651</b> | <b>0.592</b> | <b>0.576</b>   | <b>0.482</b> |
| gain of SHIFTWM (moving, mean over $k$ ): Direct +2.0 [+1.2, +2.7]; AR +4.2 [+3.3, +5.1]; 1332 win./123 ep. |             |               |              |              |                |              |
| Hamlyn                                                                                                      | Persistence | 0.742         | 0.743        | 0.700        | 0.646          | 0.569        |
|                                                                                                             | AR          | 0.774         | 0.775        | 0.746        | 0.713          | 0.665        |
|                                                                                                             | Direct      | 0.777         | 0.778        | 0.750        | 0.717          | 0.672        |
|                                                                                                             | SHIFTWM     | 0.775         | 0.777        | 0.749        | 0.716          | 0.673        |
| gain of SHIFTWM (moving, mean over $k$ ): Direct +0.1 [-0.2, +0.4]; AR +0.4 [+0.1, +0.8]; 1017 win./140 ep. |             |               |              |              |                |              |
| Bridge                                                                                                      | pending     |               |              |              |                |              |
| RT-1                                                                                                        | pending     |               |              |              |                |              |
| IWS Push-T                                                                                                  | pending     |               |              |              |                |              |
| IWS Box                                                                                                     | pending     |               |              |              |                |              |
| IWS Rope                                                                                                    | pending     |               |              |              |                |              |
| Push-T (sim)                                                                                                | pending     |               |              |              |                |              |
| Reacher                                                                                                     | pending     |               |              |              |                |              |
| Two-Room                                                                                                    | pending     |               |              |              |                |              |

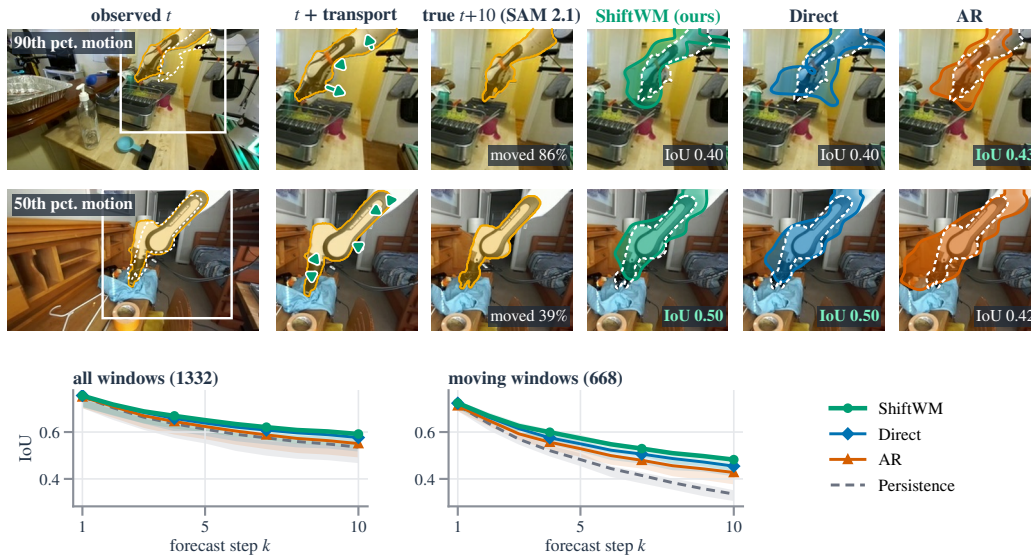


Figure 15: **Segmentation view, DROID** (target: robot arm; 1332 windows, 123 episodes). Rows: the QC-passing windows at the 90th and 50th percentile of motion. Columns as in Figure 14.

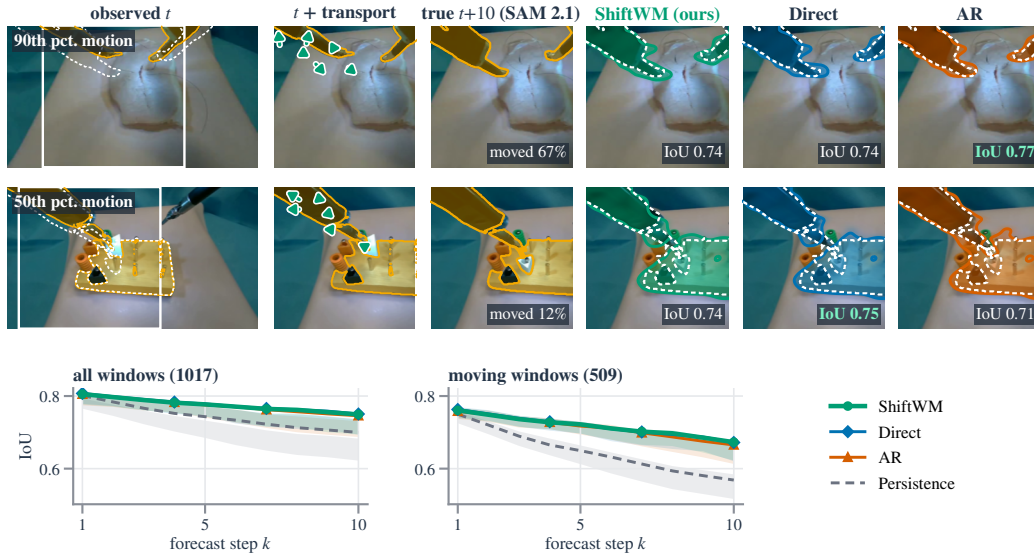


Figure 16: **Segmentation view, Hamlyn** (target: instruments; 1017 windows, 140 episodes). Rows: the QC-passing windows at the 90th and 50th percentile of motion. Columns as in Figure 14.

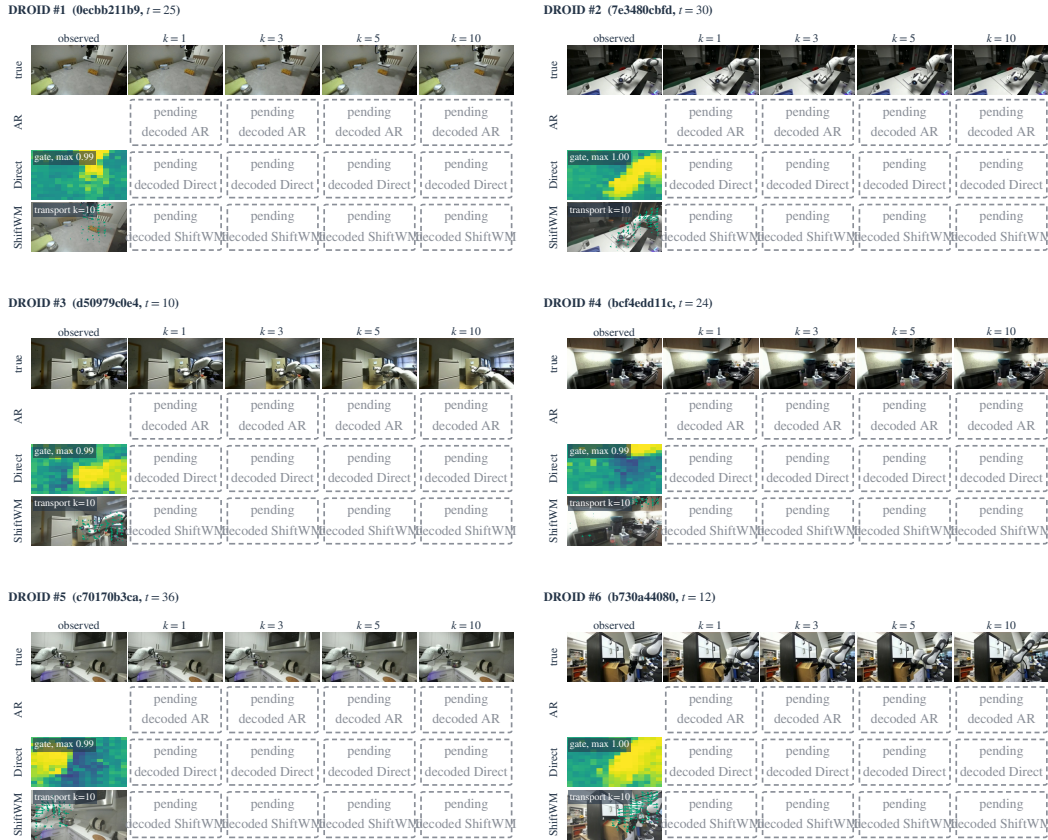
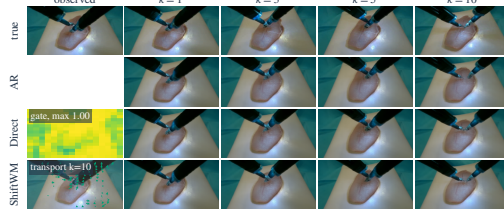
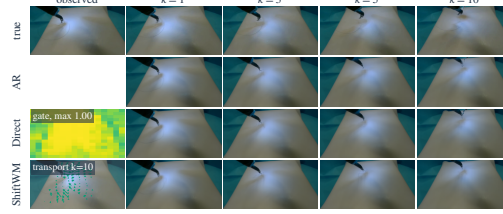


Figure 17: **DROID test rollouts (random selection).**

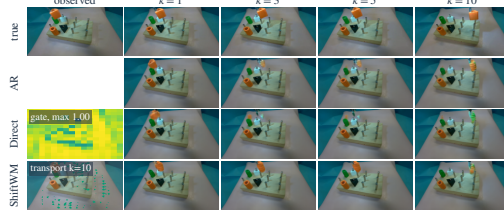
knot tying (ep 000039,  $t = 22$ )



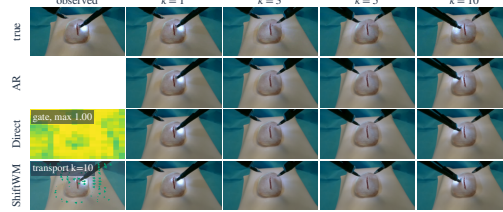
needle grasp and handover (ep 000022,  $t = 31$ )



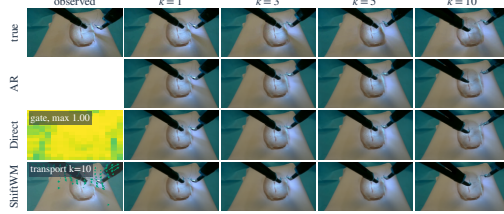
peg transfer (ep 000083,  $t = 21$ )



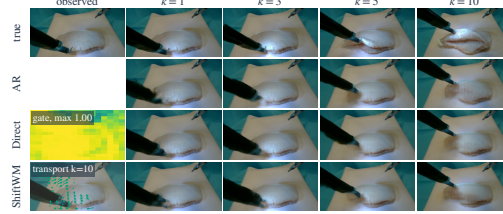
suturing 1 (ep 000042,  $t = 22$ )



suturing 2 (ep 000083,  $t = 72$ )



tissue lifting (ep 000071,  $t = 9$ )



tissue retraction (ep 000053,  $t = 9$ )

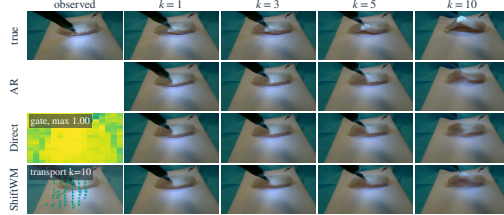


Figure 18: Surgical test rollouts (one per task, random selection).

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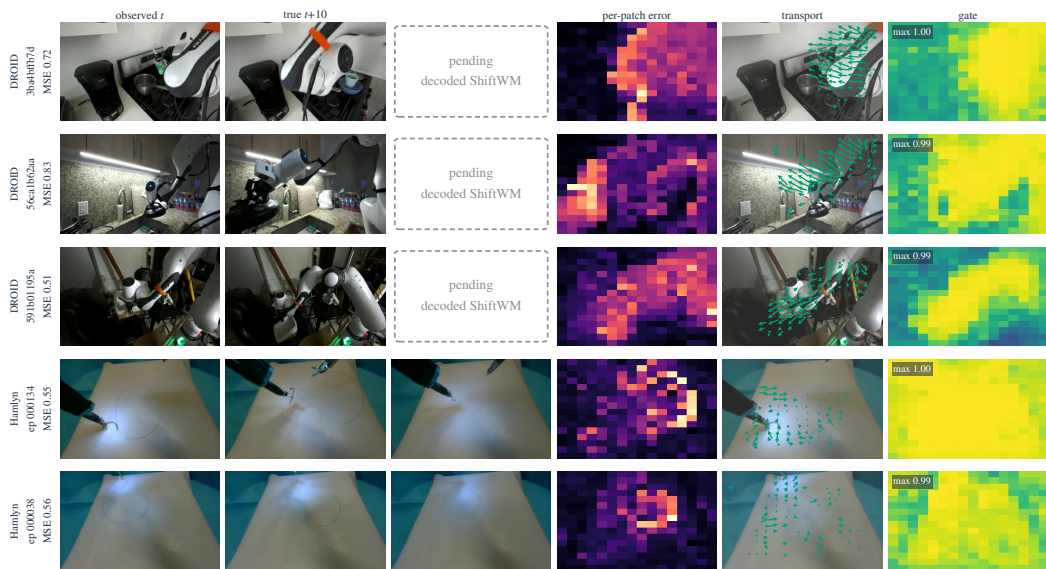


Figure 19: Failure cases.